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Inventor(s)	Seyama; Kohei

Substrate holder and bonding system

Abstract

A bonding system (90) is provided with a bonding device (80) for bonding semiconductor chips (45) to individual substrates (41), and a substrate holder (10) for holding the individual substrates (41), the bonding device (80) being provided with a stage (31) for fixing, by suction, the substrate holder (10) on a placement surface (13a), the substrate holder (10) being provided with a plate-shaped base (11) and a positioning member (21) that is provided on the base (11) and that defines the position of the individual substrates (41), the base (11) having an upper surface (11a) on which the individual substrates (41) are placed and a lower surface (11b) that is fixed by suction onto the placement surface (31a).

Inventors:	Seyama; Kohei (Tokyo, JP)
Applicant:	SHINKAWA LTD. (Tokyo, JP)
Family ID:	1000008779591
Assignee:	SHINKAWA LTD. (Tokyo, JP)
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Primary Examiner: Saad; Erin B

Attorney, Agent or Firm: JCIPRNET

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION

(1) This application is a 371 application of the International PCT application serial no. PCT/JP2020/031226, filed on Aug. 19, 2020. The entirety of the above-mentioned patent application is hereby incorporated by reference herein and made a part of this specification.

TECHNICAL FIELD

(2) The invention relates to a substrate holder holding individual substrates, a bonding system formed by the substrate holder and a bonding apparatus, and a bonding method in the bonding system.

RELATED ART

(3) In bonding of semiconductor chips, semiconductor chips are often bonded in order onto respective bonding regions of a multi-chamfering substrate in which multiple bonding regions for bonding semiconductor chips are arranged. However, in the case where dicing cannot be carried out on the substrate after bonding, or in the case where the structure is complicated like a multi-layer printed substrate and the yield is poor, a method (see, for example, Patent Document 1) in

which individual substrates are cut from the multi-chamfering substrate and placed and arranged on a palette to be conveyed to a stage of the bonding apparatus, the respective individual substrates are sucked and fixed on convex parts provided on the stage, and semiconductor chips are bonded in order thereupon.

(4) Meanwhile, in recent years, gold solder fusion bonding (see, for example, Patent Document 2) in which gold bumps are formed on multiple electrodes of an electronic component, thin solder films are formed on the surfaces of multiple copper electrodes on a substrate, and the gold in the gold bumps and the solders on the surfaces of the copper electrodes are thermal fusion bonded at the same time is adopted.

PRIOR ART DOCUMENTS

Patent Documents

(5) Patent Document 1: Japanese Laid-open No. 2000-21932 Patent Document 2: Japanese Laid-open No. 2011-254032

SUMMARY OF INVENTION

Technical Problem

(6) However, in the case where gold solder fusion bonding is adopted in bonding a semiconductor chip to a substrate as disclosed in Patent Document 2, it is required to simultaneously apply heat and a pressing load to the semiconductor chip and the substrate. However, to avoid interference with the palette, the convex part of the stage of the bonding apparatus disclosed in Patent Document 1 is smaller than the size of the individual substrate, and the individual substrate is supported in a state in which a peripheral part extends outward from the convex part. In such case, as disclosed in FIG. 9 of Patent Document 1, there is no problem if the size of the semiconductor chip is smaller than the size of the convex part. However, in recent years, there are increasing demands for bonding a semiconductor chip whose size is substantially the same as an individual substrate onto the individual substrate.

(7) At this time, in the bonding apparatus of Patent Document 1, since the individual substrate of the peripheral part is not supported by the convex part, the heat and the pressing load may not be able to be sufficiently applied to the semiconductor chip and the individual substrate of the peripheral part, and the semiconductor chip and the individual substrate may not be able to be stably bonded.

(8) Therefore, an objective of the invention is to stably bond semiconductor chips onto individual substrates.

Solution to Problem

(9) A substrate holder according to the invention holds an individual substrate to which a semiconductor chip is bonded. The substrate holder includes: a base, having a plate shape, wherein the individual substrate is placed on an upper surface of the base, and a lower surface of the base is sucked and fixed to a placement surface of a stage of a bonding apparatus; and a positioning member, provided on the base to define a position of the individual substrate placed on the upper surface of the base.

(10) Accordingly, since the entire surface of the individual substrate is held on the upper surface of the base, a semiconductor chip whose size is substantially the same as the individual substrate can be stably bonded to the individual substrate.

(11) In the substrate holder according to the invention, it may also be that in the base, a through hole penetrating in a thickness direction is provided in a region in which the individual substrate is placed, and the through hole is in communication with a suction hole provided in the stage of the bonding apparatus when the lower surface is sucked and fixed to the placement surface of the stage of the bonding apparatus.

(12) Accordingly, the individual substrate can be sucked and fixed to the upper surface of the base sucked and fixed to the placement surface of the stage.

(13) In the substrate holder according to the invention, it may also be that in the base, a plurality of

individual substrates are placed on the upper surface, and a plurality of through holes penetrating in a thickness direction are respectively provided in respective regions in which the respective individual substrates are placed, the respective through holes are in communication with at least one suction hole provided in the stage of the bonding apparatus when the lower surface is sucked and fixed to the placement surface of the stage of the bonding apparatus.

(14) Accordingly, the individual substrates can be simultaneously sucked and fixed to the upper surface of the base sucked and fixed to the placement surface of the stage.

(15) In the substrate holder according to the invention, it may also be that in the base, a groove allowing the through holes to communicate with each other is provided on the lower surface.

(16) Accordingly, the through holes can be made vacuum by using one suction hole provided on the stage to simultaneously suck and fix the respective individual substrates to the upper surface of the base.

(17) In the substrate holder according to the invention, it may also be that in the base, an upper flatness value of the upper surface and a lower flatness value of the lower surface are equal to or less than a reference flatness value, and a thickness deviation is equal to or less than a reference deviation.

(18) Accordingly, the flatness value of the upper surface when the base is sucked and fixed to the placement surface of the stage is equal to a placement flatness value of the placement surface of the stage, and the individual substrate can be held on the base at a levelness same as the case where the individual substrate is directly sucked and fixed to the placement surface of the stage.

(19) In the substrate holder according to the invention, it may also be that the positioning member is a plate-shaped member in which at least one opening with which the individual substrate is fit is formed, and is superimposed on the base so that the opening forms a concave part defining a position at which the individual substrate is placed.

(20) Accordingly, by configuring the positioning member, as a component different from the base, to be superimposed on the base, it is not required to perform processing for the concave part defining the position of the individual substrate in the base. Accordingly, the processing of the base can be simplified, and the flatness value of the base can be reduced.

(21) In the substrate holder according to the invention, it may also be that the base is a ceramic member in which the upper surface and the lower surface are polished, and the positioning member is made of metal.

(22) Accordingly, the flatness values of the upper surface and the lower surface of the base can be reduced, and the thickness deviation can be reduced. In addition, the positioning member not limited by flatness value is made by inexpensive metal. Accordingly, the cost can be reduced.

(23) A bonding system according to the invention includes: a bonding apparatus, bonding a semiconductor chip to an individual substrate; and a substrate holder, holding the individual substrate. The bonding apparatus includes a stage for sucking and fixing the substrate holder on a placement surface. The substrate holder includes: a base, having a plate shape, wherein the individual substrate is placed on an upper surface of the base, and a lower surface of the base is sucked and fixed to the placement surface of the stage; and a positioning member, provided on the base to define a position of the individual substrate placed on the upper surface of the base.

(24) In this way, since the entire surface of the individual substrate is held on the upper surface of the base, a semiconductor chip whose size is substantially the same as the individual substrate can be stably bonded to the individual substrate.

(25) In the bonding system according to the invention, it may also be that the stage includes a suction hole, in the base, a through hole penetrating in a thickness direction is provided in a region in which the individual substrate is placed, and the through hole is in communication with the suction hole provided in the stage when the lower surface is sucked and fixed to the placement surface, and sucks and fixes the individual substrate to the upper surface of the base.

(26) In this way, since the individual substrate can be sucked and fixed to the upper surface of the

base sucked and fixed to the placement surface of the stage, a semiconductor chip whose size is substantially the same as the individual substrate can be stably bonded to the individual substrate.

(27) In the bonding system according to the invention, it may also be that in the substrate holder, a plurality of individual substrates are placed on the upper surface, the positioning member respectively defines a position of each of the individual substrates placed on the upper surface of the base, the stage includes at least one suction hole, in the base, through holes penetrating in a thickness direction are respectively provided in respective regions in which the individual substrates are placed, and each of the through holes is in communication with at least one suction hole when the lower surface is sucked and fixed to the placement surface, and sucks and fixes each of the individual substrates to the upper surface of the base.

(28) Accordingly, the individual substrates can be simultaneously sucked and fixed to the upper surface of the base sucked and fixed to the placement surface of the stage to effectively perform bonding.

(29) In the bonding system according to the invention, it may also be that in the base, a groove allowing the through holes to communicate with each other is provided on the lower surface.

(30) Accordingly, the through holes can be made vacuum by using one suction hole provided on the stage to simultaneously suck and fix the respective individual substrates to the upper surface of the base.

(31) In the bonding system according to the invention, it may also be that in the base, an upper flatness value of the upper surface and a lower flatness value of the lower surface are equal to or less than a reference flatness value, and a thickness deviation is equal to or less than a reference deviation.

(32) Accordingly, the individual substrate can be held on the base at a levelness same as the case where the individual substrate is directly sucked and fixed to the placement surface of the stage, and bonding can be performed stably.

(33) In the bonding system according to the invention, it may also be that the positioning member is a plate-shaped member in which at least one opening with which the individual substrate is fit is formed, and is superimposed on the base so that the opening forms a concave part defining a position at which the individual substrate is placed.

(34) In the bonding system according to the invention, it may also be that the base is a ceramic member in which the upper surface and the lower surface are polished, and the positioning member is made of metal.

(35) A bonding method for bonding a semiconductor chip to an individual substrate according to the invention includes: a preparation step of preparing: a substrate holder, including: a base, wherein a plurality of individual substrates are placed on an upper surface of the base; and a positioning member, provided on the base to define a position of each of the individual substrates placed on the upper surface of the base, through holes penetrating in a thickness direction being respectively provided in respective regions in which the individual substrates are placed; and a bonding apparatus, including a stage having at least one suction hole, and pressing the semiconductor chip to the individual substrate to perform bonding; a placement step of placing the individual substrates on the placement surface of the substrate holder, and allowing each of the through holes to communicate with at least one of the suction hole to place the substrate holder on the placement surface of the stage; a sucking and fixing step, making the at least one of the suction hole of the bonding apparatus vacuum to suck and fix the substrate holder to the placement surface of the stage, and respectively sucking and fixing each of the individual substrates to the upper surface of the base; and a bonding process of bonding, in order, the semiconductor chip to the individual substrates.

(36) Accordingly, a semiconductor chip whose size is substantially the same as the individual substrate can be stably bonded to the individual substrate.

(37) In the bonding method according to the invention, it may also be that the bonding apparatus

includes a placement station for placing the individual substrates on the placement surface of the substrate holder, and in the placement step, after the individual substrates are placed on the placement surface of the substrate holder by using the placement station, the substrate holder on which the individual substrates are placed are conveyed onto the stage, and the substrate holder that is conveyed is placed on the placement surface of the stage.

(38) Since the individual substrates are conveyed by the substrate holder, bonding can be performed by adopting a simple method.

Effects of Invention

(39) According to the invention, a semiconductor chip can be stably bonded onto an individual substrate.

Description

BRIEF DESCRIPTION OF DRAWINGS

(1) FIG. 1 is a perspective view illustrating a bonding system according to an embodiment.

(2) FIG. 2 is a perspective view illustrating a substrate holder of the bonding system shown in FIG. 1.

(3) FIG. 3 is an exploded cross-sectional view of the substrate holder shown in FIG. 2.

(4) FIG. 4 is an elevation view illustrating a state in which a semiconductor chip is bonded to an individual substrate held in a concave part of the substrate holder of the bonding system according to the embodiment.

(5) FIG. 5 is a diagram illustrating a change of temperature of an individual substrate over time when a semiconductor chip is bonded to the individual substrate by using the bonding system shown in FIG. 1.

(6) FIG. 6 is an exploded cross-sectional view of another substrate holder.

DESCRIPTION OF EMBODIMENTS

(7) In the following, a bonding system **90** according to an embodiment is described with reference to the drawings. As shown in FIG. 1, the bonding system **90** includes a bonding apparatus **80** bonding semiconductor chips **45** onto individual substrates **41** and a substrate holder **10** holding the individual substrates **41**. In the following description, horizontal directions are set as X and Y directions, and the upper-lower direction is set as Z direction.

(8) The bonding apparatus **80** includes a stage **31** which sucks and fixes the substrate holder **10**. The upper surface of the stage **31** is a flat placement surface **31a**. On the placement surface **31**, multiple first suction holes **32** and multiple second suction holes **33** are provided. The first suction holes **32** and the second suction holes **33** are connected with a vacuum apparatus not shown herein, and suck and fix the substrate holder **10** on the placement surface **31a** by using vacuum.

(9) As shown in FIGS. 2 and 3, the substrate holder **10** is configured by using a base **11** and a positioning member **21**. The base **11** is a plate-shaped ceramic member in which the upper surface **11a** and a lower surface **11b** are polished. FIG. 2 illustrates a state in which the substrate holder **10** is set on a placement station **50** disposed in the vicinity of the bonding apparatus **80**.

(10) The positioning member **21** is a metal-made plate-shaped member formed with at least one opening **22** into which the individual substrate **41** is fit. As shown in FIG. 3, the positioning member **21** is superimposed onto the upper surface **11a** of the base **11** by providing a pin **23** extending toward the base **11** and fitting the pin **23** into a hole **14** provided in the base **11**. When the positioning member **21** is superimposed on the upper surface **11a** of the base **11**, as shown in FIG. 2, the opening **22** forms a concave part **13** defining the position at which the individual substrate **41** is placed. The concave part **13** partitions the region in which the individual substrate **41** is placed. It is noted that the hole **14** of the base **11** is larger than the pin **23** of the positioning member **21** so as to be able to absorb a thermal expansion difference between the base **11** and the positioning

member **21**.

(11) In the region of the concave part **13** of the base **11**, a through hole **12** penetrating through the base **11** in the thickness direction is provided. As shown in FIG. 4, the through hole **12** is in communication with the second suction hole **33** provided on the placement surface **31a** when the substrate holder **10** is placed on the placement surface **31a** of the stage **31**.

(12) An upper flatness value of the upper surface **11a** and a lower flatness value of the lower surface **11b** of the base **11** are each equal to or less than a reference flatness value, and a thickness deviation of the base **11** is equal to or less than a reference deviation. Here, the flatness value is a value of an interval between two flat surfaces in the case where the interval between the two flat surfaces that are parallel is minimum when a planar body is sandwiched by two geometric flat surfaces of a given length that are parallel, and is represented by using an unit of interval value (mm) between two flat surfaces (μm)/predetermined distance (mm). The reference flatness value may be set freely, and may be set, for example, between 1 (μm)/100 (mm) and 10 (μm)/100 (mm). Also, the thickness deviation is a thickness variation degree in a predetermined range, and may be set between, for example, 1 (μm)/100 (mm) and 10 (μm)/100 (mm).

(13) By defining the upper flatness value of the upper surface **11a** and the lower flatness value of the lower surface **11b** of the base **11** and the thickness deviation, the flatness value of the upper surface **11a** is equal to a placement flatness value of the placement surface **31a** of the stage **31** when the substrate holder **10** is sucked and fixed to the placement surface **31a** of the stage **31**. Therefore, the individual substrate **41** can be held on the base **11** at the same levelness as the case where the individual substrate **41** is directly sucked and fixed to the placement surface **31a** of the stage **31**.

(14) A method for bonding the semiconductor chips **45** in substantially the same size as the individual substrates **41** onto the individual substrates **41** by using the bonding system **90** configured as the above is described in the following. Firstly, the bonding apparatus **80** and the substrate holder **10** are prepared (preparation step).

(15) Firstly, as shown in FIG. 2, the substrate holder **10** is set to the placement station **50**. Then, the individual substrates **41** are placed in the concave parts **13** of the substrate holder **10** by using a collet not shown herein. Solder layers **43** as shown in FIG. 4 are formed on an electrode **42** disposed on the upper surface of each individual substrate **41**.

(16) Once all the concave parts **13** are placed with the individual substrates **41**, an underfill **44** is coated on the upper surfaces of the individual substrates **41**, and, as shown in FIG. 4, the substrate holder **10** is placed on the placement surface **31a** of the stage **31** of the bonding apparatus **80**. At this time, the lower surface **11b** of the base **11** of the substrate holder **10** covers the first suction holes **32** provided on the placement surface **31a**, and the through holes **12** of the base **11** are in communication with the second suction holes **33** provided on the placement surface **31a** (placement step).

(17) When the first suction holes **32** and the second suction holes **33** are made vacuum by using a vacuum apparatus (not shown) of the bonding apparatus **80**, the lower surface **11b** of the base **11** is sucked and held on the placement surface **31a**. In addition, when the second suction holes **33** are made vacuum, the through holes **12** of the base **11** in communication with the second suction holes **33** are also made vacuum. Accordingly, regarding the individual substrates **41** placed in the concave parts **13** of the base **11**, the entire surfaces are sucked and held on the upper surface **11a** of the base **11** in the regions of the concave parts **13**.

(18) The bonding apparatus **80** heats the stage **31** by using a heater built in the stage **31**, and, as shown in FIG. 5, the temperature of the individual substrates **41** rises to a bonding start temperature **T1**. Here, a solid line a of FIG. 5 indicates a temperature change of the individual substrate **41** in the case where the base **11** is formed by using a ceramic material with low thermal conductivity, and a dot-chain line b of FIG. 5 illustrates a temperature change of the individual substrate **41** in the case where the base **11** is formed by a ceramic material with thermal conductivity higher than

the case of the solid line a. Details regarding the solid line a and the dot-chain line b of FIG. 5 will be described subsequently.

(19) Meanwhile, the bonding apparatus **80** sucks and holds the semiconductor chip **45** at the tip of a bonding tool **36** installed to the tip of a bonding head **35**, and moves to right above the individual substrate **41**. Gold bumps **46** are formed on an electrode of the semiconductor chip **45**. A heater for heating the semiconductor chip **45** is built in the bonding head **35**.

(20) Once the temperature of the individual substrate **41** rises to the bonding start temperature **T1**, the bonding apparatus **80** rises the temperature of the semiconductor chip **45** by using the heater built in the bonding head **35**, and lowers the bonding head **35** to press the gold bumps **46** of the semiconductor chip **45** against the solder layers **43** on the electrode **42** of the individual substrate **41**. Then, as shown in FIG. 5, the bonding apparatus **80** rises the temperatures of the semiconductor chip **45** and the individual substrate **41** to a bonding temperature **T2** equal to or higher than a solder melting temperature to thermal fusion bond the gold of the gold bumps **46** and the solder layers **43** simultaneously, thereby performing gold solder fusion bonding. In addition, the melt underfill **44** is filled between the upper surface of the individual substrate **41** and the lower surface of the semiconductor chip **45**.

(21) Then, the bonding apparatus **80** conduct air into a cooling path built in the bonding head **35** to lower the temperatures of the semiconductor chip **45** and the individual substrate **41** to cure the melt metal. At this time, the underfill **44** between the upper surface of the individual substrate **41** and the lower surface of the semiconductor chip **45** is cured. Accordingly, the bonding is ended (bonding step).

(22) According to the above, in the bonding system **90** according to the embodiment, the entire surfaces of the individual substrates **41** are held by the upper surface **11a** of the base **11** in the regions of the concave parts **13**, and are sucked and fixed to the upper surface **11a** by using the through holes **12**. Therefore, even if the size of the semiconductor chip **45** and the size of the individual substrate **41** are substantially the same, heat and a pressing load can be sufficiently applied to the individual substrate **41** and the semiconductor chip **45**, and the semiconductor chip **45** can be stably bonded to the individual substrate **41**.

(23) In addition, in the base **11**, since the upper flatness value of the upper surface **11a** and the lower flatness value of the lower surface **11b** as well as the thickness deviation are defined, the individual substrates **41** can be held on the base **11** at the same levelness as the case where the individual substrates **41** are directly sucked and fixed to the placement surface **31a** of the stage **31**. Accordingly, the gold bumps **46** of the semiconductor chip **45** can be uniformly pressed onto the solder layers **43**, and favorable gold solder fusion bonding can be performed. In particular, even if the thickness of the solder layer **43** is as thin as 10 μm , favorable gold solder fusion bonding can still be performed.

(24) Moreover, the substrate holder **10** has a configuration in which the positioning member **21** made of metal is superimposed onto the base **11** as a plate-shaped ceramic member in which the upper surface **11a** and the lower surface **11b** are polished. Therefore, the flatness values and the thickness deviation of the base **11** can be reduced, and the positioning member **21** not limited by the flatness value is made by inexpensive metal. Accordingly, the cost can be reduced.

(25) In the following, the solid line a and the dot-chain line b of FIG. 5 are described. In ceramic materials, there are ceramic materials with low thermal conductivity as well as ceramic materials with higher thermal conductivity. As described above, the solid line a of FIG. 5 indicates the temperature change of the individual substrate **41** in the case where the base **11** is formed by using a ceramic material with low thermal conductivity, and the dot-chain line b of FIG. 5 illustrates the temperature change of the individual substrate **41** in the case where the base **11** is formed by a ceramic material with thermal conductivity higher than the case of the solid line a.

(26) As indicated by the solid line a of FIG. 5, in the case of the ceramic material with low thermal conductivity, even if the individual substrate **41** is heated by the heater of the stage **31**, the speed

that the heat from the stage **31** is transmitted to the individual substrate **41** is slow, and it takes time for the temperature of the individual substrate **41** to reach the bonding start temperature **T1**. However, when the semiconductor chip **45** is heated by the heater built in the bonding head **31** and the heat from the semiconductor chip **45** is transmitted to the individual substrate **41**, since the heat does not escape from the individual substrate **41** to the stage **31**, the time during which the temperature rises from the bonding start temperature **T1** to the bonding temperature **T2** is reduced. (27) Comparatively, when the base **11** is formed by using a ceramic material with thermal conductivity higher than the above, the heat transmission speed from the stage **31** to the individual substrate **41** is high, and, as indicated by the dot-chain line **b**, the time for reaching the bonding start temperature **T1** is shorter than the case of the solid line **a**. Meanwhile, since the heat escaping from the base **11** to the stage **31** in the heat flowing from the semiconductor chip **45** to the individual substrate **41** is increased, the time during which the temperature rises from the bonding start temperature **T1** to the bonding temperature **T2** is longer than the case indicated by the solid line **a**.

(28) Therefore, by preparing multiple types of the substrate holder **10** in which the thermal conductivity of the ceramic materials forming the base **11** is properly chosen, bonding matching the properties of the individual substrate **41** or the semiconductor chip **45** can be performed simply by replacing the substrate holder **10**.

(29) In the following, the configuration of a substrate holder **100** according to another embodiment is described with reference to FIG. **6**. The substrate holder **100** is configured by using a base **111** and a positioning member **121**. The positioning member **121** is a metal-made plate-shaped member formed with multiple openings **122** into which the individual substrates **41** are fit. Pins **124** protruding downward are provided at two ends of the positioning member **121**. The pins **124** are engaged with the outer surface of the base **111** to define the position of the positioning member **121** with respect to the base **111**.

(30) In multiple regions of the base **111** in which the individual substrates **41** are placed, through holes **112** respectively penetrating through an upper surface **111a** and a lower surface **111b** are provided. A groove **115** allowing the through holes **112** to laterally communicate are provided on the lower surface **111b**.

(31) As shown in FIG. **6**, when the substrate holder **100** is placed on the stage **31**, so that the groove **115** of the substrate holder **100** covers a common suction hole **34** provided on the placement surface **31a** of the stage **31**, and the common suction hole **34** is made vacuum, the respective through holes **112** become vacuum, and the base **111** can be sucked and held on the placement surface **31a** of the stage **31** at the same time when the individual substrates **41** are sucked and fixed on the upper surface **111a** of the base **111**.

(32) With such configuration, even in the case where the number of the common suction hole **34** provided on the placement surface **31a** of the stage **31** is limited or the case where the positions between the through holes **12** of the base **11** and the second suction holes **33** do not match as shown in FIG. **4**, the base **111** and the individual substrates **41** can be sucked and fixed simultaneously. Here, it suffices as long as at least one common suction hole **34** is provided on the placement surface **31a** of the stage **31**, and one or multiple suction holes **34** may be provided.

Claims

1. A substrate holder, holding an individual substrate to which a semiconductor chip is bonded, the substrate holder comprising: a base, having a plate shape, wherein the individual substrate is placed on an upper surface of the base, and a lower surface of the base is sucked and fixed to a placement surface of a stage of a bonding apparatus; and a positioning member, provided on the base to define a position of the individual substrate placed on the upper surface of the base, wherein the upper surface of the base is a flat surface, the positioning member is a plate-shaped member formed with

at least one opening with which the individual substrate is fit, and is superimposed on the upper surface of the base so that the at least one opening forms a concave part defining a position at which the individual substrate is placed, wherein on the base, a plurality of the individual substrates is placed on the upper surface, and a plurality of through holes penetrating in a thickness direction are respectively provided in respective regions in which the respective individual substrates are placed, wherein the respective through holes are in communication with at least one suction hole provided in the stage of the bonding apparatus when the lower surface is sucked and fixed to the placement surface of the stage of the bonding apparatus.

2. The substrate holder as claimed in claim 1, wherein in the base, a groove allowing the through holes to communicate with each other is provided on the lower surface.

3. A bonding system, comprising: a bonding apparatus, bonding a semiconductor chip to an individual substrate; and a substrate holder according to claim 1, holding the individual substrate, wherein the bonding apparatus comprises a stage for sucking and fixing the substrate holder on a placement surface, and the substrate holder comprises: a base, having a plate shape, wherein the individual substrate is placed on an upper surface of the base, and a lower surface of the base is sucked and fixed to the placement surface of the stage; and a positioning member, provided on the base to define a position of the individual substrate placed on the upper surface of the base, wherein the upper surface of the base is a flat surface, the positioning member is a plate-shaped member formed with at least one opening with which the individual substrate is fit, and is superimposed on the upper surface of the base so that the opening forms a concave part defining a position at which the individual substrate is placed, wherein in the substrate holder, a plurality of individual substrates are placed on the upper surface, the positioning member respectively defines a position of each of the individual substrates placed on the upper surface of the base, the stage comprises at least one suction hole, in the base, through holes penetrating in a thickness direction are respectively provided in respective regions in which the individual substrates are placed, and each of the through holes is in communication with at least one suction hole when the lower surface is sucked and fixed to the placement surface, and sucks and fixes each of the individual substrates to the upper surface of the base.

4. The bonding system as claimed in claim 3, wherein in the base, a groove allowing the through holes to communicate with each other is provided on the lower surface.

5. The substrate holder as claimed in claim 1, wherein on the base, an upper flatness value of the upper surface and a lower flatness value of the lower surface are equal to or less than a reference flatness value, and a thickness deviation is equal to or less than a reference deviation.

6. The substrate holder as claimed in claim 1, wherein the base is a ceramic member in which the upper surface and the lower surface are polished, and the positioning member is made of metal.

7. The bonding system as claimed in claim 3, wherein in the base, an upper flatness value of the upper surface and a lower flatness value of the lower surface are equal to or less than a reference flatness value, and a thickness deviation is equal to or less than a reference deviation.

8. The bonding system as claimed in claim 3, wherein the base is a ceramic member in which the upper surface and the lower surface are polished, and the positioning member is made of metal.
